

* Rank 0 tensor.

$$[M]$$

$$M = \mathbb{M}_0^{(1)}$$

* Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} M_{xx} & 0 & 0 \\ 0 & M_{xx} & 0 \\ 0 & 0 & M_{zz} \end{bmatrix}$$

$$M_{xx} = \mathbb{M}_0^{(1)} - \mathbb{M}_u^{(1)}$$

$$M_{zz} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_u^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & M_{xy} & 0 \\ -M_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xy} = \mathbb{T}_z^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} M_{xxx} & M_{xxy} & 0 \\ -M_{xxx} & -M_{xxy} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{xxy} & -M_{xxx} & 0 \end{bmatrix}$$

$$M_{xxx} = \mathbb{M}_{3a}^{(1)}$$

$$M_{xxy} = \mathbb{M}_{3b}^{(1)}$$

* Rank 3 tensor (sa). * Rank 3 tensor (a). * Rank 4 tensor (sss).

$$\begin{bmatrix} M_{xxxx} & M_{xxyy} & M_{xxzz} & 0 & 0 & 0 \\ M_{xxyy} & M_{xxxx} & M_{xxzz} & 0 & 0 & 0 \\ M_{xxzz} & M_{xxzz} & M_{zzzz} & 0 & 0 & 0 \\ 0 & 0 & 0 & M_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & M_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{M_{xxxx}}{2} - \frac{M_{xxyy}}{2} \end{bmatrix}$$

$$M_{xxxx} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_0^{(2)} + 3\mathbb{M}_{40}^{(1)} - 2\mathbb{M}_u^{(1)} - 4\mathbb{M}_u^{(2)}$$

$$M_{xxyy} = \mathbb{M}_0^{(1)} + \mathbb{M}_{40}^{(1)} - 2\mathbb{M}_u^{(1)}$$

$$M_{xxzz} = \mathbb{M}_0^{(1)} - 4\mathbb{M}_{40}^{(1)} + \mathbb{M}_u^{(1)}$$

$$M_{zzzz} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_0^{(2)} + 8\mathbb{M}_{40}^{(1)} + 4\mathbb{M}_u^{(1)} + 8\mathbb{M}_u^{(2)}$$

$$M_{yzyz} = \mathbb{M}_0^{(2)} - 4\mathbb{M}_{40}^{(1)} + \mathbb{M}_u^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & M_{xxzz} & 0 & 0 & M_{xxxy} \\ 0 & 0 & M_{xxzz} & 0 & 0 & -M_{xxxy} \\ -M_{xxzz} & -M_{xxzz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & -M_{yzzx} & 0 & 0 \\ -M_{xxxy} & M_{xxxy} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{xxzz} &= 3\mathbb{M}_u^{(3)} \\
M_{xxxy} &= 2\mathbb{T}_z^{(1)} - 2\mathbb{T}_z^{\alpha(1)} \\
M_{yzzx} &= -\mathbb{T}_z^{(1)} - 4\mathbb{T}_z^{\alpha(1)}
\end{aligned}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} M_{yzyz} & 0 & 0 \\ 0 & M_{yzyz} & 0 \\ 0 & 0 & M_{xyxy} \end{bmatrix}$$

$$\begin{aligned}
M_{yzyz} &= \mathbb{M}_0^{(3)} - 2\mathbb{M}_u^{(4)} \\
M_{xyxy} &= \mathbb{M}_0^{(3)} + 4\mathbb{M}_u^{(4)}
\end{aligned}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & M_{yzzx} & 0 \\ -M_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzzx} = -\mathbb{T}_z^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & M_{xxxy} \\ 0 & 0 & M_{xxxy} \\ 0 & 0 & M_{zzxy} \\ M_{yzyz} & M_{yzzx} & 0 \\ M_{yzzx} & -M_{yzyz} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{xxxy} &= \mathbb{T}_z^{(3)} - \mathbb{T}_z^{\alpha(2)} \\
M_{zzxy} &= \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{(4)} + 2\mathbb{T}_z^{\alpha(2)} \\
M_{yzyz} &= -3\mathbb{M}_u^{(5)} \\
M_{yzzx} &= \mathbb{T}_z^{(4)} - \mathbb{T}_z^{\alpha(2)}
\end{aligned}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & M_{yzyz} & M_{yzzx} & 0 \\ 0 & 0 & 0 & M_{yzzx} & -M_{yzyz} & 0 \\ M_{xyxx} & M_{xyxx} & M_{xyzz} & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{yzyz} &= -3\mathbb{M}_u^{(6)} \\
M_{yzzx} &= \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} \\
M_{xyxx} &= \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)} \\
M_{xyzz} &= \mathbb{T}_z^{(5)} + 2\mathbb{T}_z^{(6)} + 2\mathbb{T}_z^{\alpha(3)}
\end{aligned}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} M_{xxxx} & M_{xxyy} & M_{xxzz} & 0 & 0 & M_{xxxy} & 0 & 0 & M_{xxyx} \\ M_{xxyy} & M_{xxxx} & M_{xxzz} & 0 & 0 & -M_{xxyx} & 0 & 0 & -M_{xxxx} \\ M_{zzxx} & M_{zzxx} & M_{zzzz} & 0 & 0 & M_{zzxy} & 0 & 0 & -M_{zzxy} \\ 0 & 0 & 0 & M_{yzyz} & M_{yzzx} & 0 & M_{yzzy} & M_{yzxz} & 0 \\ 0 & 0 & 0 & -M_{yzxz} & M_{yzzy} & 0 & -M_{yzzx} & M_{yzyz} & 0 \\ -\frac{M_{xxxx}}{2} - \frac{M_{xxyy}}{2} & \frac{M_{xxyy}}{2} + \frac{M_{xxxy}}{2} & 0 & 0 & 0 & \frac{M_{xxxx}}{2} - \frac{M_{xxyy}}{2} & 0 & 0 & \frac{M_{xxxx}}{2} - \frac{M_{xxyy}}{2} \end{bmatrix}$$

$$M_{xxxx} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_0^{(2)} + 3\mathbb{M}_{40}^{(1)} - 2\mathbb{M}_u^{(1)} - 4\mathbb{M}_u^{(2)}$$

$$M_{xxyy} = \mathbb{M}_0^{(1)} + \mathbb{M}_{40}^{(1)} - 2\mathbb{M}_u^{(1)}$$

$$M_{xxzz} = \mathbb{M}_0^{(1)} - 4\mathbb{M}_{40}^{(1)} + \mathbb{M}_u^{(1)} + 3\mathbb{M}_u^{(3)}$$

$$M_{xxxy} = 2\mathbb{T}_z^{(1)} + \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)}$$

$$M_{xyyx} = 2\mathbb{T}_z^{(1)} - \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)}$$

$$M_{zzxx} = \mathbb{M}_0^{(1)} - 4\mathbb{M}_{40}^{(1)} + \mathbb{M}_u^{(1)} - 3\mathbb{M}_u^{(3)}$$

$$M_{zzzz} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_0^{(2)} + 8\mathbb{M}_{40}^{(1)} + 4\mathbb{M}_u^{(1)} + 8\mathbb{M}_u^{(2)}$$

$$M_{zzxy} = \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{(4)} + 2\mathbb{T}_z^{\alpha(2)}$$

$$M_{yzyz} = \mathbb{M}_0^{(2)} - 4\mathbb{M}_{40}^{(1)} + \mathbb{M}_u^{(2)} - 3\mathbb{M}_u^{(5)}$$

$$M_{yzzx} = -\mathbb{T}_z^{(1)} + \mathbb{T}_z^{(4)} - 4\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)}$$

$$M_{yzzy} = \mathbb{M}_0^{(2)} - 4\mathbb{M}_{40}^{(1)} + \mathbb{M}_u^{(2)} + 3\mathbb{M}_u^{(5)}$$

$$M_{yzxz} = -\mathbb{T}_z^{(1)} - \mathbb{T}_z^{(4)} - 4\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & M_{yzyz} & M_{yzzx} & 0 & M_{yzzy} & M_{yzxz} & 0 \\ 0 & 0 & 0 & M_{yzxz} & -M_{yzzy} & 0 & M_{yzzx} & -M_{yzyz} & 0 \\ M_{xyyx} & M_{xyxx} & M_{xyzz} & 0 & 0 & M_{xyxy} & 0 & 0 & -M_{xyxy} \end{bmatrix}$$

$$M_{yzyz} = \mathbb{M}_0^{(3)} - 2\mathbb{M}_u^{(4)} - 3\mathbb{M}_u^{(6)}$$

$$M_{yzzx} = -\mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)}$$

$$M_{yzzy} = -\mathbb{M}_0^{(3)} + 2\mathbb{M}_u^{(4)} - 3\mathbb{M}_u^{(6)}$$

$$M_{yzzz} = \mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)}$$

$$M_{xyyx} = \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)}$$

$$M_{xyzz} = \mathbb{T}_z^{(5)} + 2\mathbb{T}_z^{(6)} + 2\mathbb{T}_z^{\alpha(3)}$$

$$M_{xyxy} = \mathbb{M}_0^{(3)} + 4\mathbb{M}_u^{(4)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} M_{xxxx} & 0 & 0 \\ 0 & M_{xxxx} & 0 \\ 0 & 0 & M_{zzzz} \\ 0 & 0 & M_{yyzz} \\ M_{zzxx} & 0 & 0 \\ 0 & M_{xxyy} & 0 \\ 0 & M_{yyzy} & 0 \\ 0 & 0 & -\frac{M_{xxxx}}{2} + \frac{3M_{xxyy}}{2} + M_{yyzz} - M_{yzzy} + M_{zzxx} \\ \frac{M_{xxxx}}{2} - \frac{M_{xxyy}}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxx} = 3\mathbb{M}_0^{(1)} + 3\mathbb{M}_{40}^{(1)} - 2\mathbb{M}_u^{(1)} - 4\mathbb{M}_u^{(2)}$$

$$M_{zzzz} = 3\mathbb{M}_0^{(1)} + 8\mathbb{M}_{40}^{(1)} + 4\mathbb{M}_u^{(1)} + 8\mathbb{M}_u^{(2)}$$

$$M_{yyzz} = \mathbb{M}_0^{(1)} - 4\mathbb{M}_{40}^{(1)} + \mathbb{M}_u^{(1)} + 3\mathbb{M}_u^{(3)}$$

$$M_{zzxx} = \mathbb{M}_0^{(1)} - 4\mathbb{M}_{40}^{(1)} + \mathbb{M}_u^{(1)} - 3\mathbb{M}_u^{(3)}$$

$$M_{xxyy} = \mathbb{M}_0^{(1)} + \mathbb{M}_{40}^{(1)} - 2\mathbb{M}_u^{(1)}$$

$$M_{yzzy} = \mathbb{M}_0^{(1)} - 4\mathbb{M}_{40}^{(1)} + \mathbb{M}_u^{(2)} + 3\mathbb{M}_u^{(5)}$$